MIL-E-1/1333F
20 July 1976
SUPERSEDING
MIL-E-1/1333E
13 April 1972

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE

TYPE 7266

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, metal-ceramic

See figure 1

Mounting position: Any

Weight: 0.04 ounce nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Io	Ehk	TE	Barometric pressure, reduced
Unit:	V	v	ma	mAdc	V	°C	mmHg
Maximum:	6.6	600	11.0	2.2	50	75 (Note 1)	8
Minimum:	6.0	---	---	---	--	--	---
TEST CONDITIONS:	6.3	---	---	---	--	--	---

GENERAL:

Qualification - Required

(F) Denotes changes - See note 10

7266

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(F)	Qualification inspection								
(F) 1002	Barometric pressure, reduced	4	pressure = 8 +1 mmHg; voltage = 300 Vac	---	---	---	---	---	---
(F) 1031	Sweep-frequency vibration fatigue	-	Ehk = +100 Vdc; Eb = 0	---	---	---	---	---	---
	Sweep-frequency vibration fatigue test end points:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1301	Heater current	-		---	---	If	198	232	mA
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	198	232	mA
1256	Electrode current (anode)	-	Ebb = 0; Rp = 40,000 ohms	0.65	II	Ib	2	16	µAdc
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
1211	Insulation of electrodes	-	E (a-all) = -500 Vdc	0.65	II	R	10,000	---	Meg
----	Voltage drop (1)	-	Eb/Ib = 1.0 mAdc	0.65	II	Etd	0.4	2.0	Vdc
1353	Operation of rectifiers	2	Epp = 220V rms; Ehk = 0; Rℓ = 0.13 Meg; Zpp = 1,300 ohms; cℓ = 2 µF	0.65	II	---	---	---	---
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1231	Emission	-	Eb = 9 Vdc	---	---	Is	10	---	mAdc
----	Voltage drop (2)	-	Ef = 5.7 V; Eb/Ib = 1.0 mAdc	---	---	Etd	---	2.3	Vdc
1331	Direct-interelectrode capacitance	3		---	---	Cpk Chk	0.7 0.9	1.3 1.7	pF pF
----	Envelope strain	7		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
1506	Heater-cycling life	8	Ef = 7.0 V; Ehk = +70 Vdc; Eb = 0	---	---	---	---	---	---
----	Heater-cycling life test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	60	µAdc
1501	Intermittent life (1)	9	Group A; Ef = 6.3 V; Epp = 220 Vac; Ehk = -70 Vdc; Rℓ = 0.13 Meg; Cℓ = 1.0 µF;	---	---	---	---	---	---
1501	Intermittent life (2)	-	Group A; filament standby; t = 1,000 hours	---	---	---	---	---	---
----	Intermittent life-test (1) and (2) end points								
1301	Heater current	-		---	---	If	198	237	mA
----	Voltage drop (1)	-		---	---	Etd	0.3	2.5	Vdc
----	Voltage drop (2)	-		---	---	Etd	---	3.0	Vdc
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1211	Insulation of electrodes	-		---	---	R	1,000	---	Meg
1231	Emission	-		---	---	Is	7	---	mAdc
(F) 1041	Shock	5,6	Hammer angle =30°; Ehk = +100 Vdc; Eb = 0	---	---	---	---	---	---
	Shock test end points:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1301	Heater current	-		---	---	If	198	232	mA

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NOTES:

1. Operation below the rated maximum temperature is recommended for applications requiring the longest possible tube life.

2. Tap tube under test and socket assembly during operation in a rectifier circuit under conditions specified and reject for arc-backs or breakdowns. The mallet used for manual tapping shall consist of a 1/8-inch (3.18 mm) diameter fiber rod, fully inserted and glued into the small end of a No. 8 cork. The overall length of rod and cork shall be 6 inches (152.4 mm).

3. Measurement shall be made using a gounded adapter which provides shielding between external terminals of the tube.

4. The tube shall be tested as specified in method 1002 with the exception that the specified voltage shall be applied between the anode and cathode.

5. Shock test procedure and acceptance criteria shall be as follows: The tube shall be subjected to five blows in each of four positions as follows: Parallel to the cylindrical axis (two positions 180° apart); perpendicular to the cylindrical axis and parallel to the line through the heater contacts; perpendicular to the cylindrical axis and perpendicular to the line through the heater contacts.

(F)6. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

7. Tube shall be tested as specified in method 2126, except that they shall be first immersed in water at not more than 5°C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126. Tubes having high heater current of 300 mA, or more, shall be rejected as evidence of air leaks.

8. Heater-cycling life-test sampling procedure and acceptance criterial shall be as follows: First sample (n1) shall consist of 15 tubes with an acceptance number (c1) of zero. Second sample (n2) shall consist of 15 tubes with an acceptance number (cc) equal to one for the cumulative sample.

9. In a half-wave life-test circuit, the values specified for R_L and C_L may be considered as approximate and shall be adjusted initially to give not less than Io = 2.0 mAdc and ib = 20 ma with a bogey tube. The load R_L and C_L is connected in series with the anode and the supply voltage.

(F)10. Changes From Previous Issue The margins of this revision are marked with a letter F to indicate where changes from the previous revision have been made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this revision without regard to the marginal notations and the relationship to the previous issue of this document.

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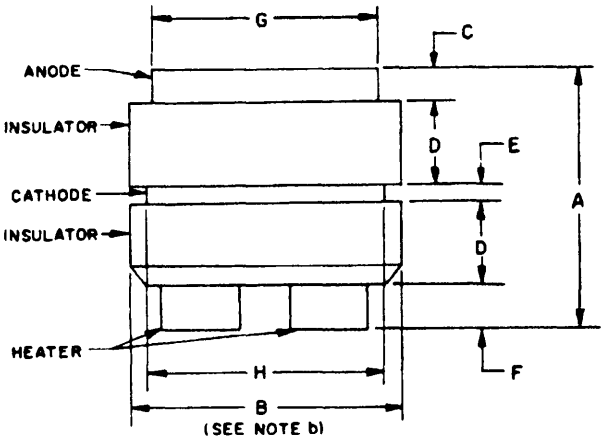
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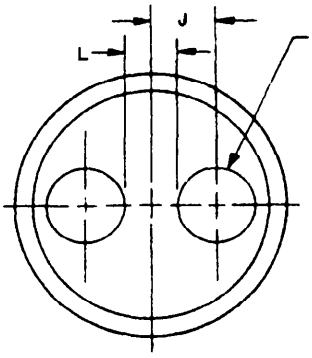
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LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (MM) IN PARENTHESES	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECTION, PART I (SEE NOTE c)		
A	.303(7.70)	.327(8.31)
B		.331(8.41)
C	.034(.86)	.046(1.17)
D	.095(2.41)	.105(2.67)
E	.022(.56)	.028(.71)
F	.042(1.07)	.058(1.47)
G	.271(6.88)	.279(7.09)
H	.281(7.14)	.289(7.34)
J	.055(1.40)	.081(2.05)
K	.086(2.18)	.094(2.39)
L	.040(1.02)	

NOTES:

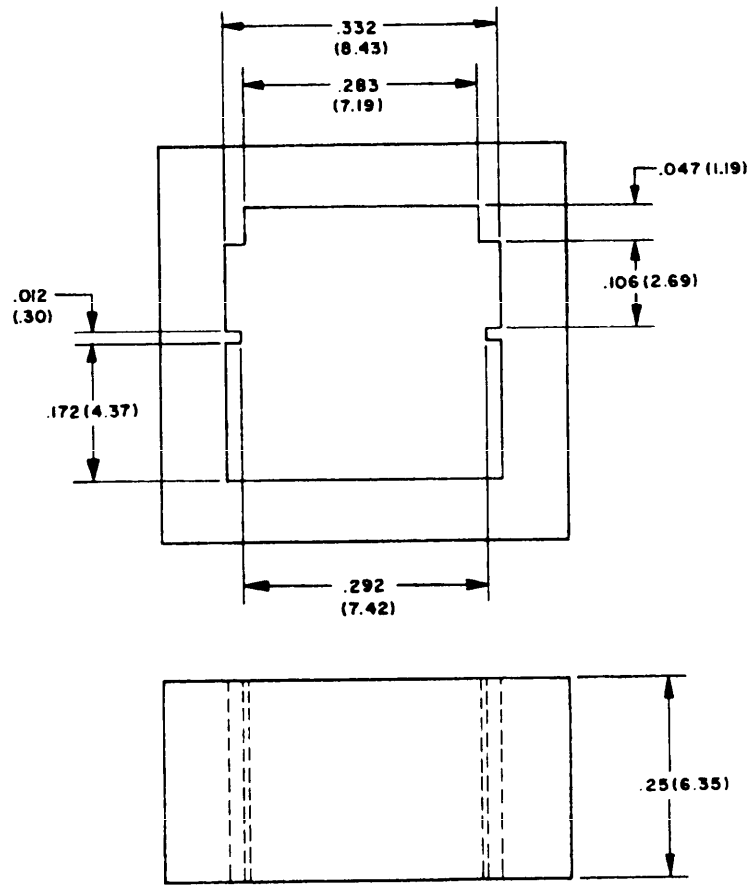
a. TUBE SHALL BE SO ALINED THAT IT IS CAPABLE OF BEING INSERTED INTO THE ALINEMENT GAGE AS SHOWN ON FIGURE 2 AND ROTATED 360 DEGREES WITHOUT BINDING.

b. DIMENSION B APPLIES TO ALL INSULATORS.

c. THE AQL FOR DIMENSIONS LISTED UNDER QUALITY CONFORMANCE INSPECTION, PART I, SHALL BE 1.0, INSPECTION LEVEL II

FIGURE 1. OUTLINE DRAWING OF ELECTRON TUBE TYPE 7266

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NOTES:

a. DIMENSIONS ARE IN INCHES.

b. METRIC EQUIVALENTS ARE IN PARENTHESES.

c. TOLERANCES OF ALL GAGE DIMENSIONS ARE ±.001 (.03 mm) UNLESS OTHERWISE SPECIFIED.

FIGURE 2 ALINEMENT GAGE

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